DIS Department of Mathematics

MYP Grade 10 Extended – Unit Test "Radicals, Exponents, Logarithms"

Name: Answers Date: ________________

You will be assessed against criterion C. You may not use your calculator for any question and you are advised to show all your working. Good luck!

Rules for logarithms:

$$\log_a xy = \log_a x + \log_a y$$

$$\log_a \frac{x}{y} = \log_a x - \log_a y$$

$$\log_a x^m = m \log_a x$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

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1. Simplify $2\sqrt{40} - 4\sqrt{3} + 1 - \sqrt{10} + \sqrt{48} - 7$ as far as possible!

$= 2\sqrt{4 \cdot 10} - 4\sqrt{3} + 1 - \sqrt{10} + \sqrt{12 \cdot 4} - 7$

$= 4\sqrt{10} - \sqrt{10} - 4\sqrt{3} - 6 + 4\sqrt{3}$

$= 3\sqrt{10} - 6$

(Total 2 marks)

2. (a) Simplify $\frac{4x^2y^9z^5}{24x^4y^2z}$

$= \frac{y^7 z^4}{6x^2}$

(1)

(b) Simplify $\frac{(2a^3)^2bc}{4a^{-4}(b^2c^5)^{-2}}$, leaving your answer with positive exponents only.

$= \frac{\cancel{4}a^6bc\,a^4(b^2c^5)^2}{\cancel{4}} = a^{10}bc \cdot b^4c^{10}$

$= a^{10}b^5c^{11}$

(2)

(Total 3 marks)

3. Write $\left(\frac{2}{3}\right)^{-2} \cdot \left(\frac{1}{2}\right)^{2} + 2$ as one fraction in simplest form.

$$= \left(\frac{3}{2}\right)^2 \cdot \frac{1}{4} + 2 = \frac{9}{4} \cdot \frac{1}{4} + \frac{32}{16}$$
$$= \frac{41}{16}$$

(Total 2 marks)

4. (a) Write $8^{-\frac{2}{3}}$ as a fraction in simplest form.

$$= (2^3)^{-\frac{2}{3}} = 2^{-2} = \frac{1}{4}$$

(1)

(b) Write $64^{\frac{5}{6}}$ as an integer number.

$$= (8^2)^{5/6} = 8^{\frac{5}{3}} = (2^3)^{\frac{5}{3}} = 2^5 = \underline{32}$$

(1)
(Total 2 marks)

5. Expand and simplify as far as possible.

(a) $(1 - 3\sqrt{6})(1 - \sqrt{6})$

$$= 1 - 4\sqrt{6} + 18 = \underline{19 - 4\sqrt{6}}$$

(2)

(b) $(1 - 3\sqrt{2})^3$

$$= (1-3\sqrt{2})^2(1-3\sqrt{2})$$
$$= (1 - 6\sqrt{2} + 18)(1-3\sqrt{2})$$
$$= (19 - 6\sqrt{2})(1-3\sqrt{2}) = 19 - 57\sqrt{2} - 6\sqrt{2} + 36$$
$$= 55 - 63\sqrt{2}$$

(2)
(Total 4 marks)

6. Solve the linear equation below. Write your solution in simplest form with no radicals or negatives in the denominator.

$5\sqrt{3}x - \sqrt{7} = \sqrt{7} + 1 - 2x$

Note: The "x" on the left hand side of the equation is **not** underneath the square root.

$$5\sqrt{3}x + 2x = 2\sqrt{7} + 1$$

$$x(5\sqrt{3} + 2) = 2\sqrt{7} + 1$$

$$x = \frac{(2\sqrt{7}+1)}{(5\sqrt{3}+2)} \cdot \frac{(5\sqrt{3}-2)}{(5\sqrt{3}-2)} = \frac{10\sqrt{21} - 4\sqrt{7} + 5\sqrt{3} - 2}{75 - 4}$$

$$= \frac{10\sqrt{21} - 4\sqrt{7} + 5\sqrt{3} - 2}{71}$$

(Total 3 marks)

7. Express the following with a **positive** integer denominator. Simplify as far as possible.

(a) $\frac{7}{\sqrt{4}} = \frac{7}{2}$

(1)

(b) $\frac{1}{\sqrt{7}-2} \cdot \frac{\sqrt{7}+2}{\sqrt{7}+2} = \frac{\sqrt{7}+2}{3}$

(2)

(c) $\frac{2-3\sqrt{5}}{2\sqrt{5}+1} \cdot \frac{(2\sqrt{5}-1)}{(2\sqrt{5}-1)} = \frac{4\sqrt{5} - 2 - 30 + 3\sqrt{5}}{20 - 1}$

$$= \frac{7\sqrt{5} - 32}{19}$$

(2)

(Total 5 marks)

8. Solve the following equations.

(a) $\log_2(2x) = 3$

$$2^3 = 2x \rightarrow \underline{x = 4}$$

(1)

(b) $\log_{\frac{1}{2}}\left(\frac{1}{32}\right) = x$

$$\left(\frac{1}{2}\right)^x = \frac{1}{32} \qquad \therefore \underline{x = 5}$$

(1)

(c) $2^{1-\frac{x}{2}} = \frac{1}{8}$

$$2^{1-\frac{x}{2}} = 2^{-3} \rightarrow 1 - \frac{x}{2} = -3$$

$$2 - x = -6$$

$$\therefore \underline{x = 8}$$

(2)

(d) $\log_4(x-1) - \log_4(2x+1) + 1 = 0$

$$\log_4\left(\frac{x-1}{2x+1}\right) = -1$$

$$\therefore \frac{1}{4} = \frac{x-1}{2x+1} \rightarrow \cancel{2x}+1 = \cancel{4x}-4$$

$$5 = 2x$$

$$\therefore \underline{x = \frac{5}{2}}$$

(2)

(e) $\ln(3x-2)+\ln\left(\frac{1}{x}\right)=-2$

Give your answer in the form $x=\frac{a}{b-e^{-2}}$, where $a, b \in \mathbb{N}$.

$$\ln\left(\frac{3x-2}{x}\right)=-2$$

$$\therefore e^{-2}=\frac{3x-2}{x} \to xe^{-2}=3x-2$$

$$xe^{-2}-3x=-2$$

$$x(e^{-2}-3)=-2$$

$$\therefore x=\frac{-2}{e^{-2}-3}=\underline{\underline{\frac{2}{3-e^{-2}}}}$$

(2)

(f) $\log_9(x)+2\log_x(9)=3$

$$\log_9 x+\log_x 9^2=3$$

$$\log_9 x+\frac{\log_9 81}{\log_9 x}=3 \qquad \text{let } a=\log_9 x$$

$$a+\frac{2}{a}=3$$

$$a^2+2=3a$$

$$\therefore a^2-3a+2=0$$

$$(a-2)(a-1)=0$$

$$\therefore a=1 \text{ or } 2$$

(3)

(Total 11 marks)

$$\therefore \underline{\underline{x=9 \text{ or } 81}}$$

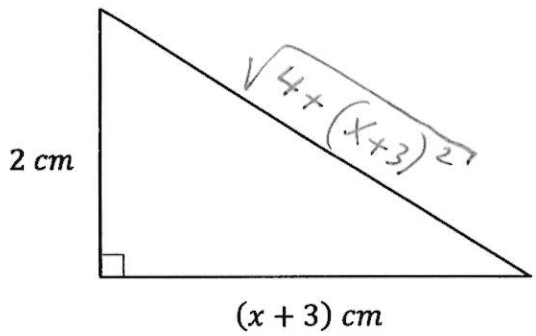
9. Find the value of $\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \ldots \cdot \log_{31} 32$.

e.g. $= \frac{\cancel{\log 3}}{\log 2} \cdot \frac{\cancel{\log 4}}{\cancel{\log 3}} \cdot \frac{\cancel{\log 5}}{\cancel{\log 4}} \cdot \ldots \cdot \frac{\cancel{\log 31}}{\cancel{\log 30}} \cdot \frac{\log 32}{\cancel{\log 31}}$

$= \frac{\log 32}{\log 2} = \log_2 32 = \underline{\underline{5}}$

(Total 3 marks)

10. The diagram below (**not** drawn to scale) shows a right triangle with side lengths 2 *cm* and $(x+3)$ *cm*.



(a) Find, in terms of x, an expression for the perimeter of this triangle.

$p = 2 + (x+3) + \sqrt{4+(x+3)^2}$

$= 5 + x + \sqrt{4+(x+3)^2}$

(2 marks)

(b) Hence, given that the perimeter is 5 cm, find the value of x.

$\cancel{5} = \cancel{5} + x + \sqrt{4+(x+3)^2}$

$-x = \sqrt{\ldots}$

$x^2 = 4 + (x+3)^2$

$\cancel{x^2} = 4 + \cancel{x^2} + 6x + 9 \rightarrow x = \underline{\underline{-\frac{13}{6}}}$

(2 marks)
(Total 4 marks)